



How to Create Personalized User Guides with DITA



Building Your First Taxonomy for Classification of Software Documentation

HOW TO CREATE A TAXONOMY? WHAT IS
METADATA AND TAXONOMY?



1. Why building a product taxonomy?
2. A few important terms to know.
3. How to build a taxonomy?

Why Building a Product Taxonomy?



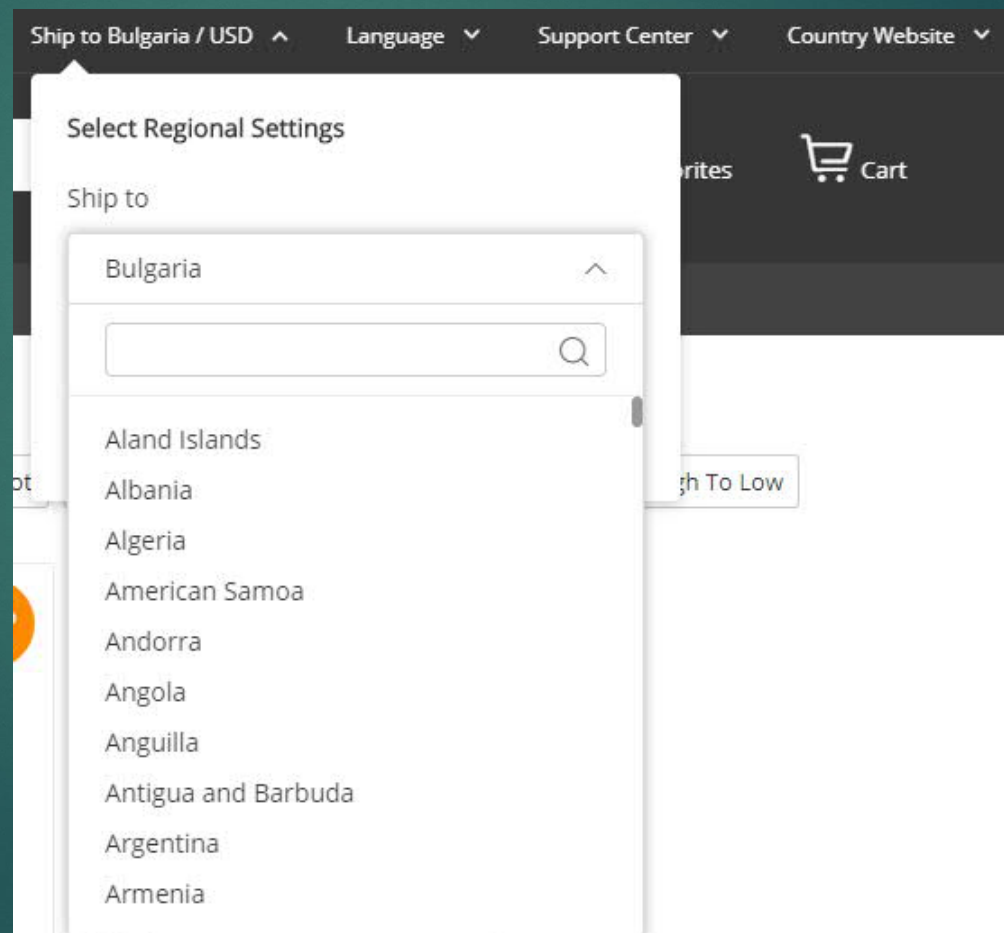
Achieve a Common Understanding

Huh! I only see the
Sun and some rocks.
And fog!

It's it a lovely sunrise,
honey?!



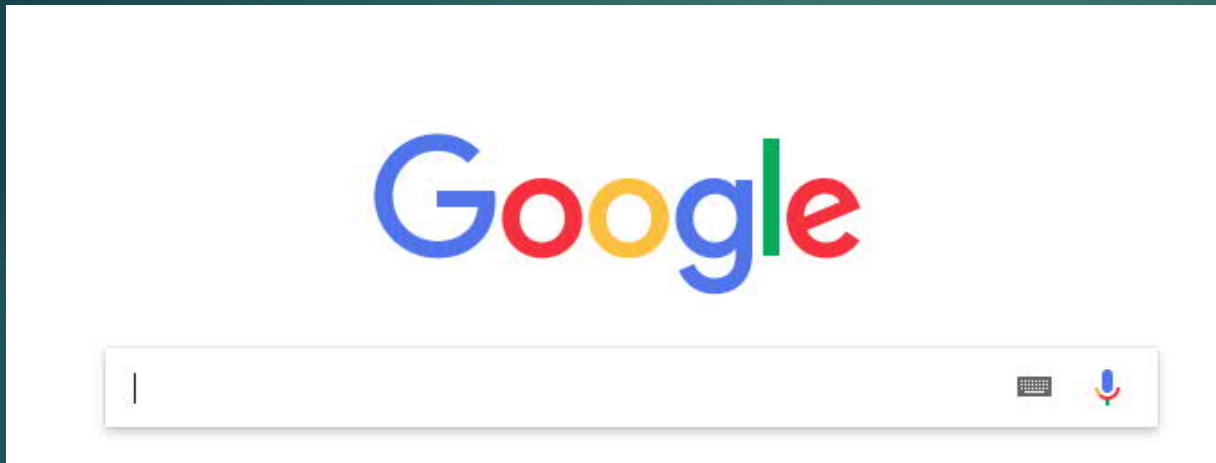
Define a Controlled Vocabulary



Aggregate Different Products Based on Common Characteristics



People Search Behavior – Generic Search vs. Faceted Search



SHOP BY DEPARTMENT ▼

< All Categories

Bags & Shoes (2058)

- Men's Bags
- Women's Bags
- Coin Purse & Card Holder
- Men's Shoes
- Women's Shoes
- Waist Packs
- Wallets

Narrow Search Results

☐ ON SALE	319
☐ IN STOCK	2019
☐ CLEARANCE	180

Listed

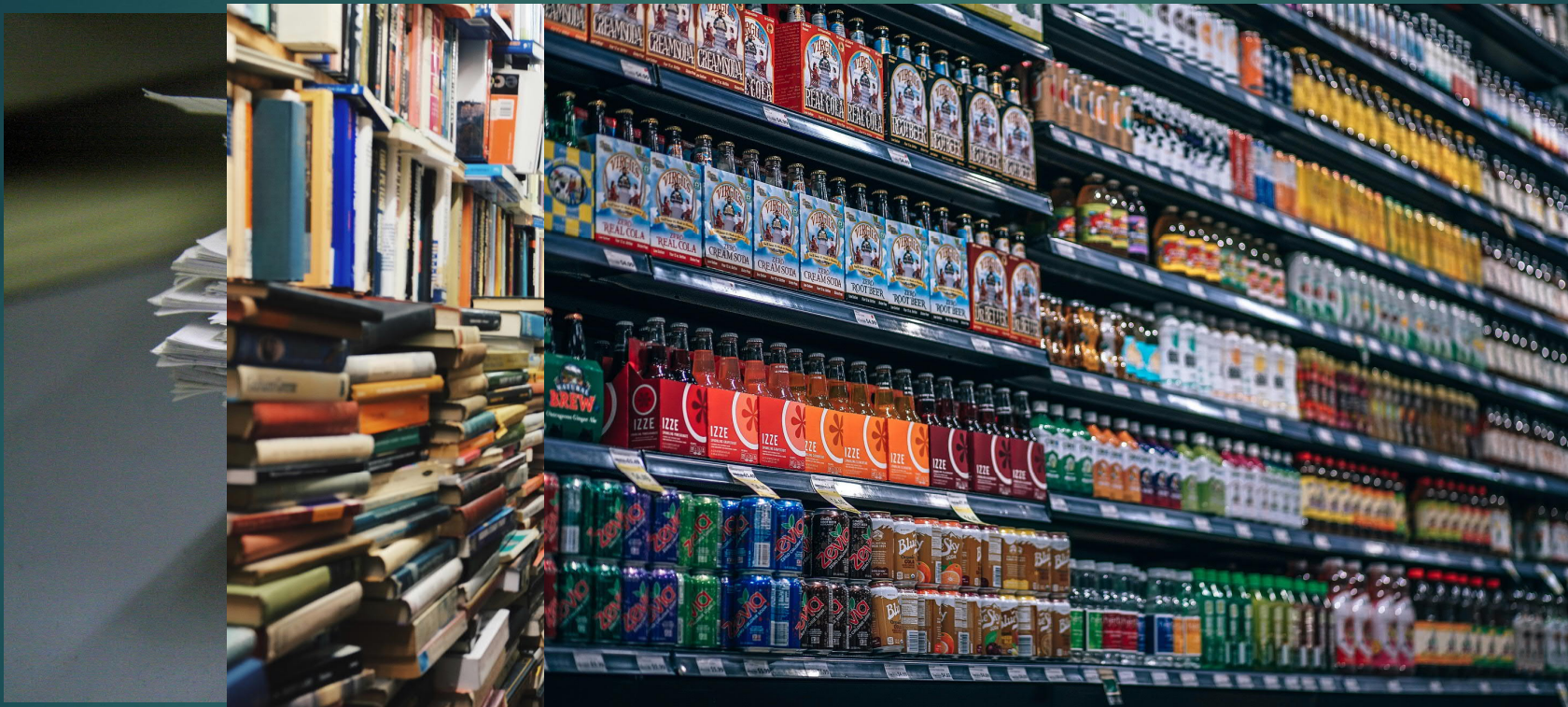
☐ Last 7 Days	78
☐ Last 14 Days	391
☐ Last 30 Days	556
☐ Last 90 Days	763

Price Range

My Story



Think About Your Story!





Terminology & Definitions

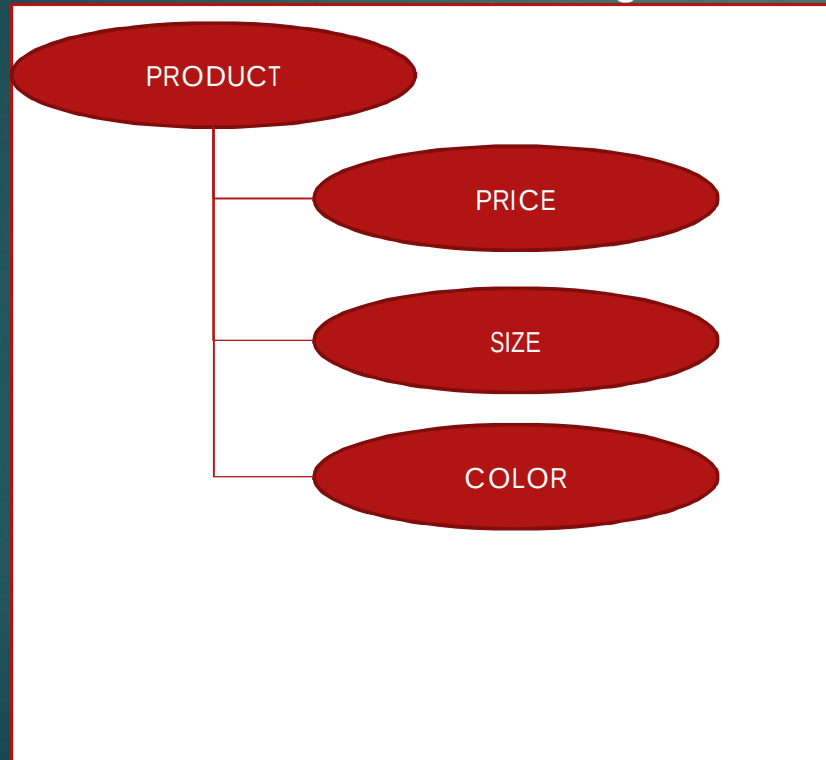
Terminology & Definitions: Metadata

Metadata: simply data about data!



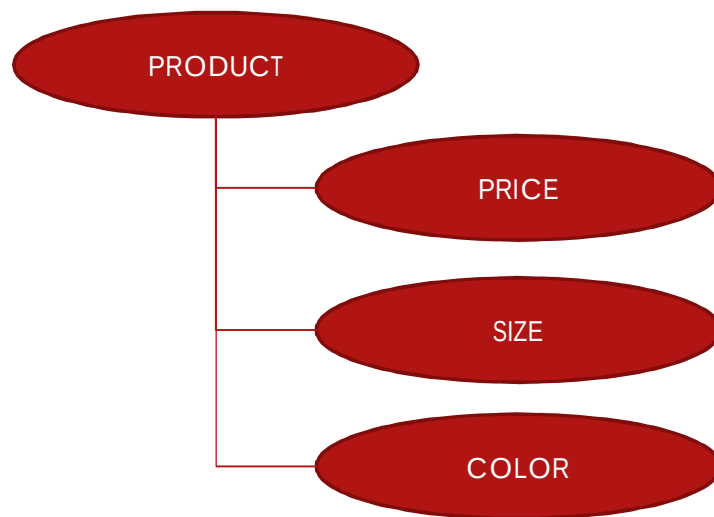
Terminology & Definitions: Taxonomy

Taxonomy: a system for naming and organizing things, especially plants and animals, into groups that share similar qualities. Mathematically, a hierarchical taxonomy is a tree structure of classifications for a given set of objects.



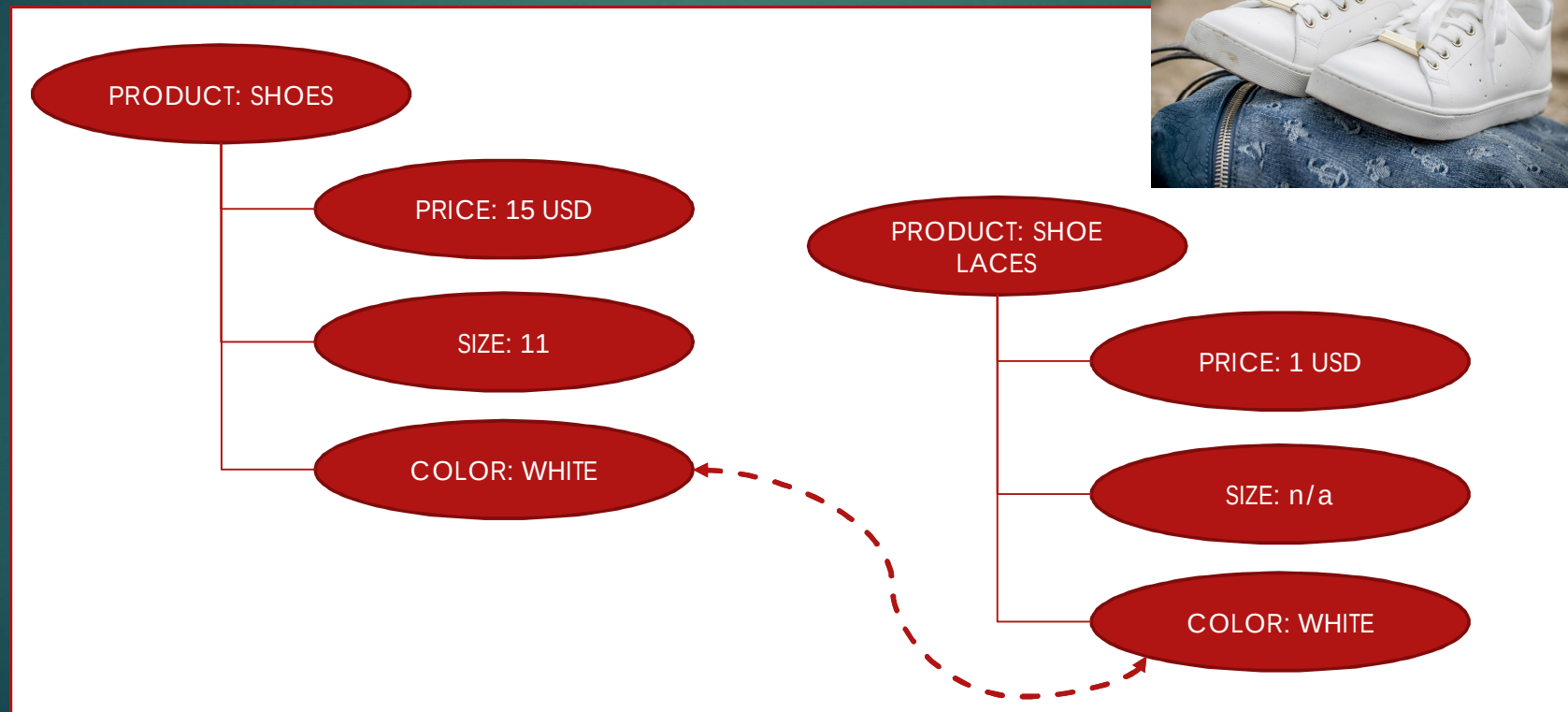
Terminology & Definitions: Classification

Classification: the act or process of dividing things into groups according to their type



Terminology & Definitions: Ontology

Ontology: a set of concepts and categories in a subject area or domain that shows their properties and the relations between them.



Strategies

1. Build taxonomy using a standard
2. Build taxonomy by description
3. Build taxonomy by comparison



Strategy 1: Build Taxonomy Using a Standard

- ▶ https://en.wikipedia.org/wiki/Product_classification
- ▶ https://en.wikipedia.org/wiki/Metadata_standard
- ▶ <https://www.google.com/basepages/producttype/taxonomy.en-US.txt>

```
# Google_Product_Taxonomy_Version: 2015-02-19
```

```
Animals & Pet Supplies
Animals & Pet Supplies > Live Animals
Animals & Pet Supplies > Pet Supplies
Animals & Pet Supplies > Pet Supplies > Bird Supplies
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Cage Accessories
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Cage Accessories > Bird Cage Bird Baths
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Cage Accessories > Bird Cage Food & Water Dishes
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Cages & Stands
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Food
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Gyms & Playstands
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Ladders & Perches
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Toys
Animals & Pet Supplies > Pet Supplies > Bird Supplies > Bird Treats
```

Name	Focus	
DDI	Archiving and Social Science	The Data Documentation Initiative (DDI) is a metadata specification, which is written in XML.
EBUCore	The EBUCore metadata set for audiovisual content	EBUCore is a set of descriptive and administrative metadata for audiovisual content, developed by the European Broadcasting Union (EBU). EBUCore is largely used as a standard for metadata exchange between broadcasters.
CCDM		The EBU Class Conceptual Data Model (CCDM) is a conceptual data model for audiovisual content, developed by the European Broadcasting Union (EBU). CCDM is used as a standard for metadata exchange between broadcasters.
FOAF		The Friend of a Friend (FOAF) project is a project to develop a standard for representing people and their relationships on the Web.
EAD		Encoded Archival Description (EAD) is a standard for describing archival collections, developed by the International Council on Archives (ICA).
CDD		Categories for the Description of Works (CDD) is a standard for describing works, developed by the International Council on Archives (ICA).
VRA		Visual Resources Association – the Visual Resources Association (VRA) is a standard for describing visual resources, developed by the Visual Resources Association (VRA).
DC		The Darwin Core is a metadata specification for biodiversity data, developed by the Darwin Core Consortium.
OIE		Online Information Exchange - international (OIE) is a standard for exchanging information online, developed by the International Organization for Standardization (ISO).
LOM	Education	The main purpose of the Common Learning Objects Metadata (LOM) is to provide a standard for describing learning objects.
CSDGM	Geographic data	Ecological Metadata Language is a standard for describing ecological data, developed by the Ecological Metadata Language Consortium.
		Content Standard for Digital Geospatial Metadata (CSDGM) is a standard for describing geospatial data, developed by the International Geospatial Standards Organization (IGSO).

Exercise 1

Using a web search, each of you find a standard list of values, showing the counties in the world.

Share your findings with your partner.



Strategy 2: Build a Taxonomy by Description

1. Write down as detailed description as possible.
2. Underline the category-value pairs you have identified.
3. Construct the taxonomy.



Example 2



This is a candy stick.

The candy stick product has 3 parts: the candy, the stick and the package of the candy.

The candy has the shape of a circle, with a diameter of 2.5 cm by 0.4 cm. Its weight is 3 grams. It is made of normal melted sugar. The color of the candy is yellow.

The stick has straight shape and has a length of 4 cm. It is suitable for food products.

The package is made of 100% transparent plastics with square form. Each side is 3 cm long. There is a stamping with the barcode on it. The package has an image of a crying out of happiness face.

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Exercise 2: Build a Taxonomy by Description

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3. Construct the taxonomy.



Example 2

Share with your partner the taxonomy you have created.

Strategy 3: Build Taxonomy by Comparison

1. Take two or more products of the same kind.
2. Write down what are their common characteristics.
3. Write down their differences.
4. Construct the taxonomy by the categories you have identified.



Exercise 3: Build Taxonomy by Comparison

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Example 3

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The stick has straight shape and has a length of 4 cm. It is suitable for food products.

The package is made of 100% transparent plastics with square form. Each side is 3 cm long. There is a stamping with the barcode on it. The package has an image of a angry face.

Example 3

Share with the class the taxonomy you have created.

Lessons Learned and Wrap up

- ▶ Metadata is simply data about data.
- ▶ You can take a taxonomy home! It is just a file, a description of an object.
- ▶ To build the taxonomy, start by searching for an existing standard first!
- ▶ Describe the product in details and identify the possible taxonomy terms.
- ▶ Compare your taxonomy with other, similar product to identify the gaps in the taxonomy.
- ▶ Have fun while doing that and bring value to your customer!





Governance of Profiling Values

Why?

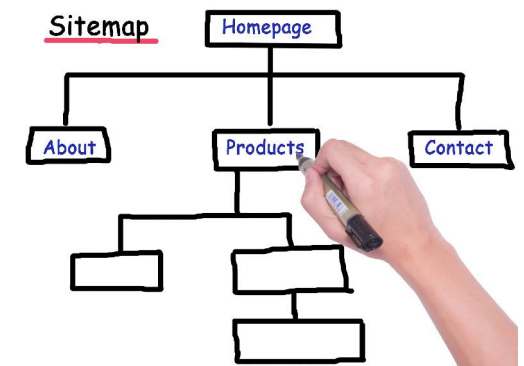
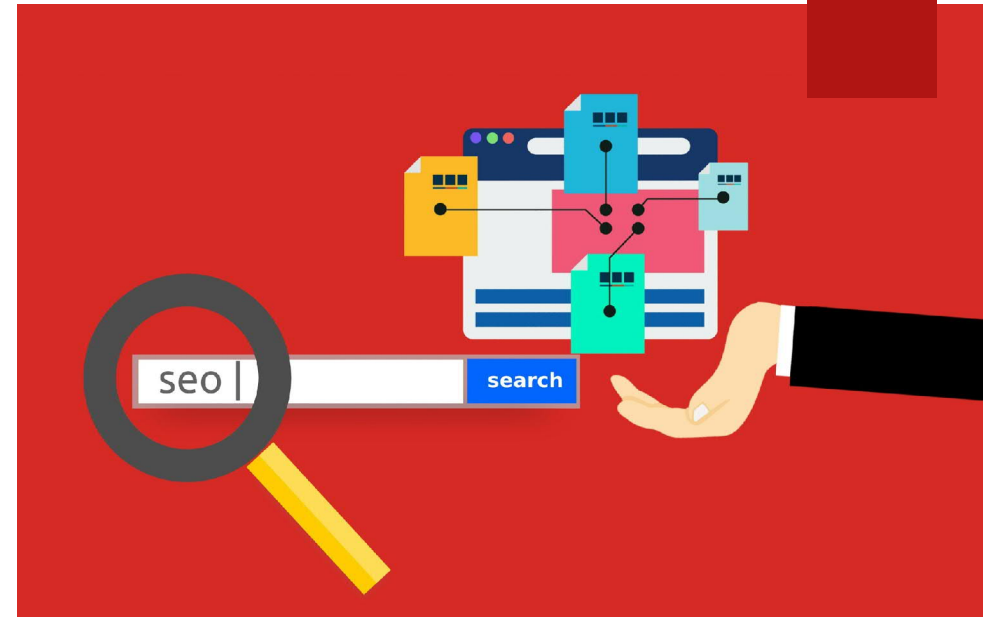
- ▶ If you need to govern and pre-define the list of profiling values to be used in the scope of a DITA map.
- ▶ If you need to ensure that the author use precise list of values from a large set of profiling values.
- ▶ If you want to extend and modify the list of profiling values, without specialization effort.
- ▶ If you want to build a hierarchy of profiling values.

Subject Scheme Maps in DITA

Subject scheme maps can be used to define controlled values and subject definitions.

The controlled values can be bound to attributes, as well as element and attribute pairs.

The subject definitions can contain metadata and provide links to more detailed information; they can be used to classify content and provide semantics that can be used in taxonomies and ontologies.



Creating a Subject Scheme Map



Prerequisite: define a taxonomy of terms to use for profiling.

For example, the way living beings are divided up into:

- ❑ Kingdom,
- ❑ Phylum,
- ❑ Class,
- ❑ Order,
- ❑ Family,
- ❑ Genus,
- ❑ Species.

Or your company products, databases, and so on.

Preferences

Filtered

Editor / Edit Modes / Author / Profiling/Conditional Text

Editor

- Edit Modes
 - Author
 - Profiling/Conditional Text
 - Colors and Styles
 - Attributes Rendering

To import the profiling attributes defined in a set of DITAVAL files and create profiling condition sets based on these files, click "Import DITAVAL".

Profiling Attributes

Document type	Attribute	Values
DocBook	vernum	vernum
DocBook*	wordsize	32bit
DocBook 4	lang	en;fr
DocBook 5	xml:lang	en;fr
DocBook 5	audience	expert
DITA	biology	kingdom

☒ Report invalid profiling attribute values (DITA)

☒ Allow contributing extra profiling attribute values

[Configure profiling colors and styles](#)

Profiling Condition Sets

A condition set is a collection of profiling attributes

Document type	Name	Condition
DITA	include audience ...	audience
DITA	Include only Nov...	audience

☒ Show excluded content

Profiling Attribute

Note: Define only profiling attributes that actually exist in your schema. This feature does not add profiling support to XML vocabularies that do not have it already.

Document type: DITA

Attribute name: biology

Display name: biology

Value	Label	Description
kingdom		

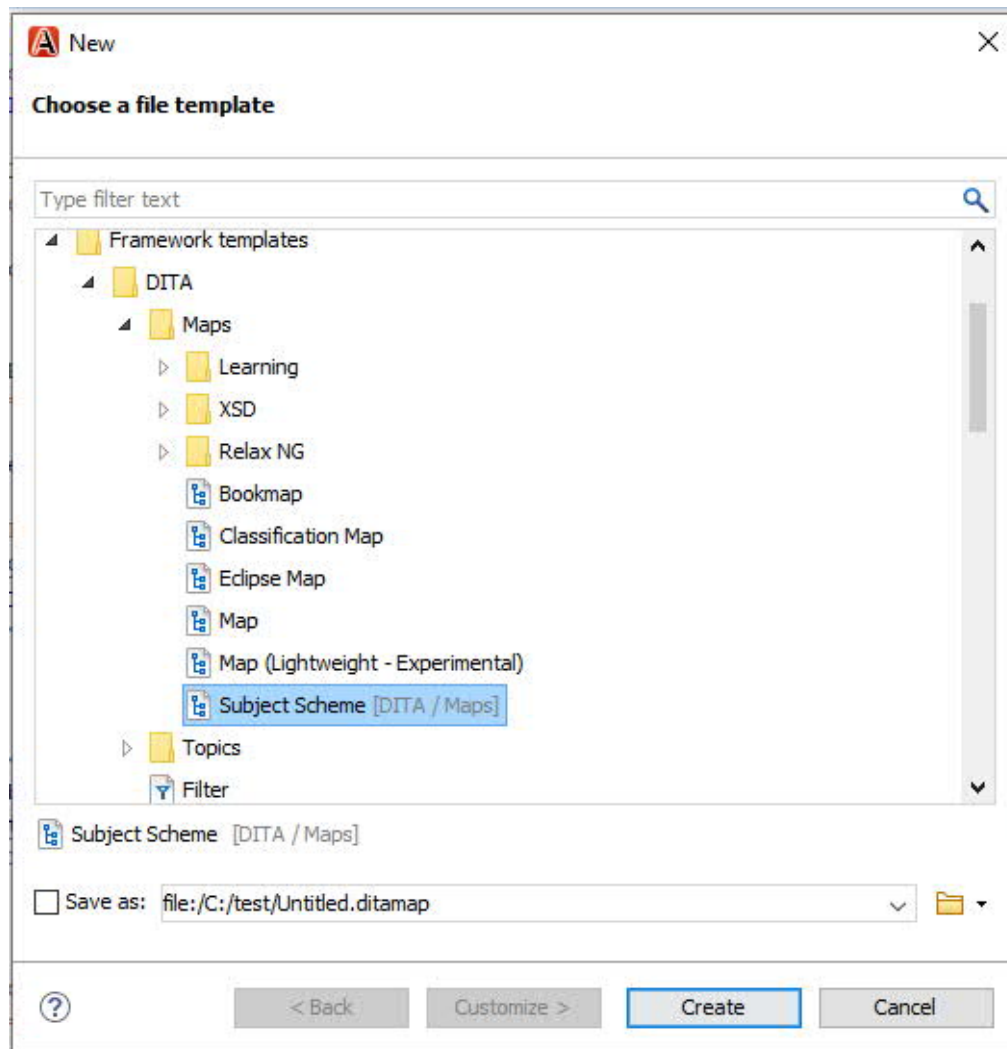
New Edit Delete

☐ Single value

☒ Multiple values separated by: <space>

OK Cancel

Step 1: Create the Profiling Attribute



Step 2: Create the Subject Scheme Map

Step 3: Add new or modify the subjects

```
<!-- Define the profiling category we want to create or overwrite -->
<subjectdef keys="biology">
  <!-- Define the profiling values in this category -->
  <subjectdef keys="kingdom">
    <subjectdef keys="phylum">
      <subjectdef keys="class">
        <subjectdef keys="order">
          <subjectdef keys="family">
            <subjectdef keys="genus">
              <subjectdef keys="species"> </subjectdef>
            </subjectdef>
          </subjectdef>
        </subjectdef>
      </subjectdef>
    </subjectdef>
  </subjectdef>
</subjectdef>
</subjectdef>
</subjectdef>
</subjectdef>
```

Step 4: Bind the newly defined values to the attribute

```
<enumerationdef>  
  <attributedef name="biology"/>  
  <subjectdef keyref="biology"/>  
</enumerationdef>
```


Step 5: Remove the obsolete attributes

```
<enumerationdef>  
  <attributedef name="platform"/>  
  <subjectdef/>  
</enumerationdef>
```

Step 5: Reference the Subject Scheme Map

- ▶ A DITA map reference a subject scheme map via <mapref>.
- ▶ Make sure the type is subjectScheme:

```
<mapref  
  href="subject_scheme.ditamap"  
  format="ditamap"  
  type="subjectScheme"  
>
```

DEMO

